Experiment:38

Design a C program to simulate SCAN disk scheduling algorithm.

Code: -

```
#include <stdio.h>

#include <stdlib.h>

int main()

{

   int n, head, seek_time = 0;

   int disk_size;

   printf("Enter the disk size: ");

   scanf("%d", &disk_size);

   printf("Enter the number of disk requests: ");

   scanf("%d", &n);

   int request_queue[n];

   printf("Enter the disk request queue:\n");


   for (int i = 0; i < n; i++)

   {

      scanf("%d", &request_queue[i]);

   }


   printf("Enter the initial position of the disk head: ");

   scanf("%d", &head);

   for (int i = 0; i < n - 1; i++)

   {

      for (int j = i + 1; j < n; j++)
```

```
    {
      if (request_queue[i] > request_queue[j])
      {
        int temp = request_queue[i];
        request_queue[i] = request_queue[j];
        request_queue[j] = temp;
      }
    }
  }

  printf("\nSCAN (Elevator) Disk Scheduling:\n");
  printf("Head Movement Sequence: %d", head);
  int index = 0;
  while (index < n && request_queue[index] < head)
  {
    index++;
  }
  for (int i = index; i < n; i++)
  {
    seek_time += abs(head - request_queue[i]);
    head = request_queue[i];
    printf(" -> %d", head);
  }
  if (head != disk_size - 1)
  {
    seek_time += abs(head - (disk_size - 1));
```

```c
            head = disk_size - 1;
            printf(" -> %d", head);
        }
        for (int i = index - 1; i >= 0; i--)
        {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];

            printf(" -> %d", head);
        }
        printf("\nTotal Seek Time: %d\n", seek_time);
        printf("Average Seek Time: %.2f\n",
               (float)seek_time / n);
        return 0;
}
```

Output:

```
Enter the disk size: 5
Enter the number of disk requests: 3
Enter the disk request queue:
3 6 9
Enter the initial position of the disk head: 3

SCAN (Elevator) Disk Scheduling:
Head Movement Sequence: 3 -> 3 -> 6 -> 9 -> 4
Total Seek Time: 11
Average Seek Time: 3.67


...Program finished with exit code 0
Press ENTER to exit console.
```